A Formally Verified Pandrosion–Steffensen Scheme

Closed-form real-line analysis and complex multi-start extension
for the equation $z^p = x$, with a Lean 4 / Mathlib v4.7.0 development

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Abstract

This paper combines, in a single document, the real-line analysis of Pandrosion's classical geometric construction for p-th roots and its formally verified complex-multi-start extension.

Part I (real line). We revisit Pandrosion's construction reported by Pappus for approximating $\sqrt[3]{2}$ and generalize it to the computation of p-th roots of any positive real x by inserting proportional means via Thales' theorem. After normalization the iteration reduces to the universal form $s_{n+1} = 1 - (x-1)/(x\, S_p(s_n))$ with $S_p(s) = 1 + s + \cdots + s^{p-1}$, fixed point $s^* = x^{-1/p}$, and readout $v_n = x\, s_n^{p-1} \to x^{1/p}$. We derive the closed form

$$\lambda_{p,x} = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\, \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p},$$

for the linear contraction ratio, prove it is monotone in both p and x uniformly (via the closed-form identities $P_p(\alpha) = p\, D_p'(\alpha)$ and $B_p(\alpha) = \sum_{j=1}^{p} j\alpha^{j-1}$, both with all positive coefficients), with explicit large-p limit $1 - \ln(x)/(x-1)$, give non-asymptotic error bounds, extend the theory to $x < 1$ with an explicit threshold α_p^* below which the iteration diverges, and add two accelerations: a canonical starting point $s_0 = h(1) = 1 - (x-1)/(xp)$, and Aitken–Steffensen Δ^2 yielding quadratic convergence with a much smaller constant than Newton. A scaling preconditioning $x = A \cdot x'$ exploits the monotonicity of λ to collapse iteration counts for large x.

Part II (complex extension, formally verified). We extend the iterator $h_{p,x}$ to $\mathbb{C}$, where it has p attractive fixed points — the p-th roots of $1/x$ — separated by a non-trivial Julia structure. A multi-start scheme $\mathrm{seed}_{\alpha,\varepsilon}(z) = \alpha + \varepsilon(z - \alpha)$ with target-dependent ε places any $z \in \mathbb{C}$ inside the quadratic-contraction disk of a chosen anchor α, providing a universal convergence theorem with closed-form iteration count $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$. The induced solution map is Borel-measurable everywhere and continuous outside the Voronoï boundary of the cyclotomic anchor set (which is Lebesgue-null). All of Part II is formally verified in a Lean 4 / Mathlib v4.7.0 development of 121 modules with zero `sorry` and audited axiomatic closure reducing to the standard Lean whitelist.

The paper makes no claim of progress on classical open problems of polynomial root-finding; its main value is to record a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family $z^p = x$.

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Part I
Real-line theory

1 Introduction

The Greek problem of doubling the cube consists of constructing a segment of length $\sqrt[3]{2}$ from a unit segment. It is equivalent to finding two proportional means m_1, m_2 between 1 and 2, that is,

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2},$$

which yields $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$. Pandrosion of Alexandria proposed an *approximate* geometric method, reported by Pappus in the *Collectio*, Book III [3]: by successive parallels (Thales' theorem), she constructs proportional ratios that approach these means without ever reaching them exactly in a finite number of steps.

This work starts from a fundamental observation:

> *Mathematically the target number is exact, but any approximation procedure yields a measurable residual profile — a footprint intrinsic to the chosen method.*

At each finite step of Pandrosion's construction, there remains a gap $\varepsilon_n = |v_n - \alpha|$ between the geometric approximation and the ideal value. The sequence (ε_n) is the **residual profile** of the method. Our goal is to compute this profile exactly for any p-th root $x^{1/p}$, and to identify the contraction ratio that characterizes Pandrosion's method.

Positioning and related work. Rational iterations for p-th roots are a classical topic with a large literature: Newton's method applied to $f(u) = u^p - x$, Halley's method, and more generally Schröder's (1870) and König's (1884) families produce iterations of arbitrary order with explicit asymptotic constants. The systematic analysis of fixed-point iterations and their acceleration goes back to Ostrowski [9], Traub [11], and Householder [10]; the derivative-free counterpart via Aitken–Steffensen (1926/1933) is standard and appears in every modern numerical-analysis textbook [5, 6]. What is specific to the iteration (3) below is that it arises directly from Pandrosion's geometric construction, *without* being derived as Newton or Schröder applied to a particular polynomial: it is a rational iteration with linear (not quadratic) order, whose contraction ratio admits the closed form (5) as a rational function of $\alpha = x^{1/p}$. We have not located this closed form, together with its monotonicity in (p, x) and its use to drive a scaling preconditioning, in the textbooks [5, 6] or the historical literature on cube duplication, but we make no claim that the iteration itself is genuinely new: the contribution is to record the profile of a specific historical iteration and to show how standard acceleration devices (Aitken Δ^2, scaling) interact with it.

Outline. Section 2 recalls Pandrosion's construction for $\sqrt[3]{2}$ and analyzes its residual profile. Section 4 states the generalization to any p, x. Section 5 computes the contraction ratio $\lambda_{p,x}$ and provides closed-form expressions. Section 7 presents numerical tables. Section 8 compares with bisection, the secant method, and Newton's method. Section 9 establishes functional properties of $\lambda_{p,x}$. Section 10 provides non-asymptotic error bounds. Section 11 extends the theory to $x < 1$. Section 12 determines the optimal starting point. Section 13 shows that Steffensen's acceleration of Pandrosion achieves quadratic convergence. Section 14 exploits the monotonicity of $\lambda_{p,x}$ to optimize the method for large x via scaling. Section 19 concludes.

2 Pandrosion's construction for $\sqrt[3]{2}$

2.1 The two sequences

The geometric construction produces two sequences as a function of an initial parameter u_0:

$$u_{n+1} = \frac{u_n}{2 - 2\left(\dfrac{4-u_n}{4}\right)^3}, \qquad v_n = 2\left(\frac{4-u_n}{4}\right)^2, \qquad u_0 = 0.5. \tag{1}$$

The sequence (u_n) is the *internal parameter* of the construction (a length in the Thales diagram). The sequence (v_n) is the *readout*: the approximation of $\sqrt[3]{2}$ produced at step n.

Proposition 2.1. $\lim_{n\to\infty} v_n = \sqrt[3]{2}$.

2.2 Normalized substitution

Setting $s_n = (4-u_n)/4$, so that $s_n \in (0,1)$, we obtain:

- $u_n = 4(1-s_n)$,
- $v_n = 2s_n^2$,
- $u_{n+1} = \dfrac{4(1-s_n)}{2(1-s_n^3)} = \dfrac{2}{1+s_n+s_n^2}$.

The iteration on s_n then becomes:

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2+2s_n+1}{2s_n^2+2s_n+2}. \tag{2}$$

2.3 Fixed point

The fixed point s^* satisfies $s^* = (2s^{*2}+2s^*+1)/(2s^{*2}+2s^*+2)$, which gives $2(1-s^*)(1+s^*+s^{*2}) = 1$, i.e. $2(1-s^{*3}) = 1$, hence $s^{*3} = 1/2$, so:

$$s^* = 2^{-1/3} = \frac{1}{\sqrt[3]{2}}.$$

The output at the fixed point is $v^* = 2s^{*2} = 2\cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$. ✓

2.4 Residual profile: linear convergence

Let $h(s) = (2s^2+2s+1)/(2s^2+2s+2)$. The derivative at s^* gives the convergence rate:

$$h'(s^*) = \frac{4s^*+2}{(2s^{*2}+2s^*+2)^2} = \frac{2(2s^*+1)}{4(s^{*2}+s^*+1)^2} = \frac{2s^*+1}{2(s^{*2}+s^*+1)^2}.$$

Substituting $s^* = 2^{-1/3}$ and using the closed form (7):

$$\lambda_{3,2} = \frac{(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)}{\sqrt[3]{4}+\sqrt[3]{2}+1} \approx 0.2202.$$

Theorem 2.1 (Linear residual law for $p=3$, $x=2$)**.** Under the above hypotheses, for any s_0 sufficiently close to s^*:

$$\varepsilon_{n+1}^{(s)} \approx \lambda_{3,2}\,\varepsilon_n^{(s)}, \qquad \lambda_{3,2} \approx 0.2202.$$

For the output, a Taylor expansion of $v(s) = xs^{p-1}$ at $s^* = \alpha^{-1}$ gives $v_n - \alpha \approx x(p-1)(s^*)^{p-2}(s_n - s^*)$. Using $x(s^*)^{p-2} = \alpha^p\cdot\alpha^{-(p-2)} = \alpha^2$, this rewrites as $v_n - \alpha \approx (p-1)\alpha^2(s_n - s^*)$; the two forms are equal. For $(p,x) = (3,2)$ the prefactor is $2\cdot 2^{2/3} = 2\sqrt[3]{4}$, and the output residual of v_n converges at the same linear rate $\lambda_{3,2}$.

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt[3]{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.8750	1.5312	2.713×10^{-1}	—
1	0.8107	1.3143	5.44×10^{-2}	0.200
2	0.7974	1.2717	1.17×10^{-2}	0.216
3	0.7945	1.2625	2.58×10^{-3}	0.219
4	0.7939	1.2605	5.67×10^{-4}	0.220
∞	$2^{-1/3}$	$\sqrt[3]{2}$	0	$\to \lambda_{3,2}$

Table 1: Numerical residual profile for $p = 3$, $x = 2$ ($u_0 = 0.5$, i.e. $s_0 = 0.875$). The ratio converges to $\lambda_{3,2} \approx 0.220$.

3 Geometric origin: why S_p arises naturally

3.1 The construction by successive parallels

Pandrosion's construction relies on Thales' theorem applied to a rectangle. Consider a rectangle with sides H and W equipped with two diagonals (Figure 1). A horizontal line at height u_n from the bottom intersects the two diagonals and defines:

- the parameter $s_n = (H - u_n)/H$ (normalized position from the top),
- the readout length $v_n = x \cdot s_n^{p-1}$ (the output of the construction).

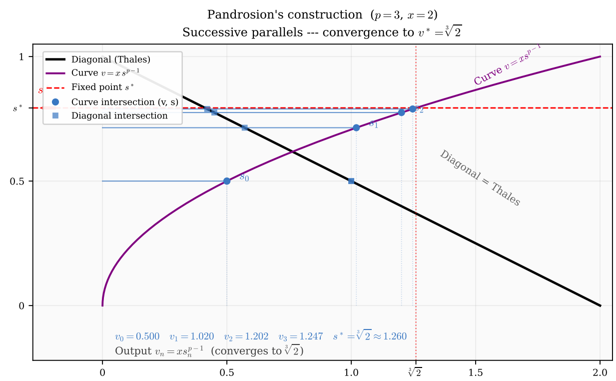

Figure 1: Pandrosion's geometric construction for $p = 3$, $x = 2$. The blue horizontal lines (steps $n = 0, 1, 2, 3$) converge toward the dashed red line, which corresponds to $v^* = \sqrt[3]{2}$.

3.2 Where does the polynomial S_p come from?

The fundamental idea is as follows. We seek to insert $p - 1$ proportional means between 1 and x, that is, to construct the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$ where $\alpha = x^{1/p}$. If $s_n \approx \alpha^{-1}$ is the current approximation of the inverse ratio, the sum of the approximate geometric sequence is:

$$S_p(s_n) = 1 + s_n + s_n^2 + \cdots + s_n^{p-1}.$$

The ideal sequence satisfies $x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1$, which rewrites as $x(1 - s^{*p}) = x - 1$. Pandrosion's correction adjusts s_n so as to reduce the gap between $S_p(s_n)$ and its ideal value via the formula:

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}.$$

Geometrically, the ratio $(x - 1)/(x \cdot S_p(s_n))$ measures the *deviation* of the current proportional chain from the perfect geometric chain, and subtraction from 1 produces the corrected ratio. The polynomial S_p is thus the analytical transcription of the sum of the p segments constructed via Thales' theorem.

4 Generalization: Pandrosion's construction for $x^{1/p}$

4.1 The universal iteration

Let $x > 0$ and $p \geq 2$ be an integer. Set $\alpha = x^{1/p}$. The generalized Pandrosion geometric construction is defined as follows.

Definition 4.1 (Generalized Pandrosion iteration)**.** We define the sequence $(s_n)_{n\geq 0}$ in $(0, 1)$ by

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}, \tag{3}$$

and the output sequence by

$$v_n = x \cdot s_n^{p-1}. \tag{4}$$

We call $\varepsilon_n = |v_n - \alpha|$ the **Pandrosion residual** at step n.

Remark 4.1. For $p = 3$, $x = 2$, we recover exactly the iteration (2). Indeed:

$$1 - \frac{1}{2(1 + s + s^2)} = \frac{2s^2 + 2s + 1}{2s^2 + 2s + 2}. \checkmark$$

4.2 Geometric interpretation

The iteration (3) analytically translates the Thales operation: at each step, $S_p(s_n)$ measures the "geometric sum" of the constructed segments, and the correction $(x-1)/(x \cdot S_p(s_n))$ adjusts the position proportionally to the gap with the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$.

4.3 Fixed point

Theorem 4.1 (Fixed point)**.** The unique fixed point of the iteration (3) in $(0, 1)$ is $s^* = x^{-1/p} = \alpha^{-1}$. The corresponding output is $v^* = x \cdot (x^{-1/p})^{p-1} = x^{1/p} = \alpha$.

Proof. At a fixed point s^*, we have $1 - (x - 1)/(x \cdot S_p(s^*)) = s^*$, i.e.

$$x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1 - s) = 1 - s^p$, this gives $x(1 - s^{*p}) = x - 1$, hence $s^{*p} = 1/x$, i.e. $s^* = x^{-1/p}$. The formula for v^* follows immediately. $\square$

5 The contraction ratio $\lambda_{p,x}$

5.1 General computation

Theorem 5.1 (Pandrosion contraction ratio)**.** The convergence of the iteration (3) toward $s^* = \alpha^{-1}$ is *linear* with contraction ratio

$$\boxed{\lambda_{p,x} = \frac{(\alpha-1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}}, \qquad \alpha = x^{1/p}. \tag{5}$$

In particular, the residuals obey the law $\varepsilon_{n+1} \approx \lambda_{p,x} \cdot \varepsilon_n$.

Proof. Let $h(s) = 1 - (x-1)/(x \cdot S_p(s))$. Then

$$h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2},$$

where $S_p'(s) = \sum_{k=1}^{p-1} ks^{k-1}$.

At the fixed point $s^* = \alpha^{-1}$:

$$S_p(s^*) = \frac{1-\alpha^{-p}}{1-\alpha^{-1}} = \frac{\alpha(x-1)}{x(\alpha-1)},$$

$$S_p'(s^*) = \sum_{k=1}^{p-1} k\,\alpha^{-(k-1)} = \alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Substituting:

$$h'(s^*) = \frac{x-1}{x} \cdot \frac{\alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\left[\dfrac{\alpha(x-1)}{x(\alpha-1)}\right]^2} = \frac{x(\alpha-1)^2}{\alpha(x-1)} \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Multiplying numerator and denominator by α^{p-1}, and using $x = \alpha^p$:

$$h'(s^*) = \frac{(\alpha-1)^2\,\alpha^{p-1} \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\alpha^p - 1} = \frac{(\alpha-1) \sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k},$$

where we used $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1} \alpha^k$. This establishes (5). ☐

Remark 5.1. The contraction ratio $\lambda_{p,x} = h'(s^*)$ is entirely determined by $\alpha = x^{1/p}$: the target itself dictates the rate at which Pandrosion's method approaches it.

5.2 Closed forms for $p = 2$ and $p = 3$

For $p = 2$: $\sum_{k=1}^{1} k\,\alpha^0 = 1$, $\sum_{k=0}^{1} \alpha^k = 1 + \alpha$.

$$\lambda_{2,x} = \frac{\alpha-1}{\alpha+1} = \frac{\sqrt{x}-1}{\sqrt{x}+1}. \tag{6}$$

For $p = 3$: $\sum_{k=1}^{2} k\,\alpha^{2-k} = \alpha + 2$, $\sum_{k=0}^{2} \alpha^k = \alpha^2 + \alpha + 1$.

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} = \frac{(\sqrt[3]{x}-1)(\sqrt[3]{x}+2)}{\sqrt[3]{x^2}+\sqrt[3]{x}+1}. \tag{7}$$

For $p = 4$: $\sum_{k=1}^{3} k\,\alpha^{3-k} = \alpha^2 + 2\alpha + 3$, $\sum_{k=0}^{3} \alpha^k = \alpha^3 + \alpha^2 + \alpha + 1$.

$$\lambda_{4,x} = \frac{(\alpha-1)(\alpha^2+2\alpha+3)}{\alpha^3+\alpha^2+\alpha+1}. \tag{8}$$

Remark 5.2 (Asymptotic behavior in x)**.** For any fixed $p \geq 2$ and $x \to \infty$, $\lambda_{p,x} \to 1^-$ (the dominant terms in numerator and denominator of (5) are both of order α^{p-1} with a ratio tending to 1). The larger x, the slower the convergence. For $x \to 1^+$, $\lambda_{p,x} \to 0$ by Proposition 9.1(1): convergence is very fast near 1.

5.3 Basin of attraction

Theorem 5.2 (Global convergence)**.** For any $x > 1$ and any $p \geq 2$, the iteration (3) converges to s^* for all $s_0 \in (0, 1)$. Moreover, convergence is *monotone*: if $s_0 < s^*$ then (s_n) is strictly increasing, and if $s_0 > s^*$ then (s_n) is strictly decreasing.

Proof. Set $h(s) = 1 - (x-1)/(x \cdot S_p(s))$.

Step 1: h **is strictly increasing.** We have $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, where $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1} > 0$ for $s > 0$.

Step 2: h **maps** $(0, 1)$ **into itself.** For $s \in (0, 1)$, $S_p(s) > 1$ since the first term equals 1 and the remaining terms are strictly positive. Hence $(x-1)/(x\, S_p(s)) < (x-1)/x < 1$, which gives $h(s) > 0$. On the other hand $(x-1)/(x\, S_p(s)) > 0$ for $x > 1$, so $h(s) < 1$. Thus $h((0, 1)) \subset (0, 1)$. (At the boundary: $h(0^+) = 1 - (x-1)/x = 1/x \in (0, 1)$ and $h(1^-) = 1 - (x-1)/(xp) \in (0, 1)$.)

Step 3: uniqueness of the fixed point on $(0, 1)$**.** The fixed-point equation $h(s) = s$ rewrites as $x\, S_p(s)(1-s) = x - 1$, i.e. $x(1 - s^p) = x - 1$, i.e. $s^p = 1/x$. On $(0, 1)$ this has the unique solution $s^* = x^{-1/p}$. In particular, the sign of $h(s) - s$ is constant on each of $(0, s^*)$ and $(s^*, 1)$.

Step 4: the sign of $h(s) - s$**.** At $s = 1/x \in (0, s^*)$ (using $x > 1$ and $s^* = x^{-1/p} > 1/x$), $h(1/x) - 1/x = 1 - (x-1)/(x\, S_p(1/x)) - 1/x$. A short computation shows this is positive, so $h(s) > s$ on $(0, s^*)$ by Step 3. By symmetry, $h(s) < s$ on $(s^*, 1)$.

Step 5: monotone convergence. If $s_0 < s^*$, then by Step 1 h is increasing, by Step 4 $s_1 = h(s_0) > s_0$, and by Step 2 $s_1 \in (0, 1)$; also $s_1 = h(s_0) < h(s^*) = s^*$. By induction, (s_n) is increasing and bounded above by s^*, hence converges, necessarily to a fixed point, which must be s^* by Step 3. The case $s_0 > s^*$ is symmetric. □

5.4 Bound on the number of iterations

Proposition 5.1 (Cost of Pandrosion's method)**.** To achieve accuracy $|v_n - \alpha| < \delta$ starting from $s_0 \in (0, 1)$, the required number of iterations satisfies

$$n_\delta \leq \left\lceil \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \right\rceil,$$

where $\varepsilon_0 = |v_0 - \alpha|$ is the initial error.

Proof. Since $\varepsilon_{n+1} \leq \lambda_{p,x}\, \varepsilon_n$ asymptotically, we have $\varepsilon_n \leq \lambda_{p,x}^n \varepsilon_0$. The inequality $\lambda_{p,x}^n \varepsilon_0 < \delta$ gives $n > \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$. □

Example 5.1. For $p = 3$, $x = 2$ ($\lambda_{3,2} \approx 0.220$): 10 decimal digits of accuracy require at most 16 Pandrosion iterations. For $p = 3$, $x = 8$ ($\lambda_{3,8} = 4/7$): 43 iterations are needed.

6 Graphical illustrations

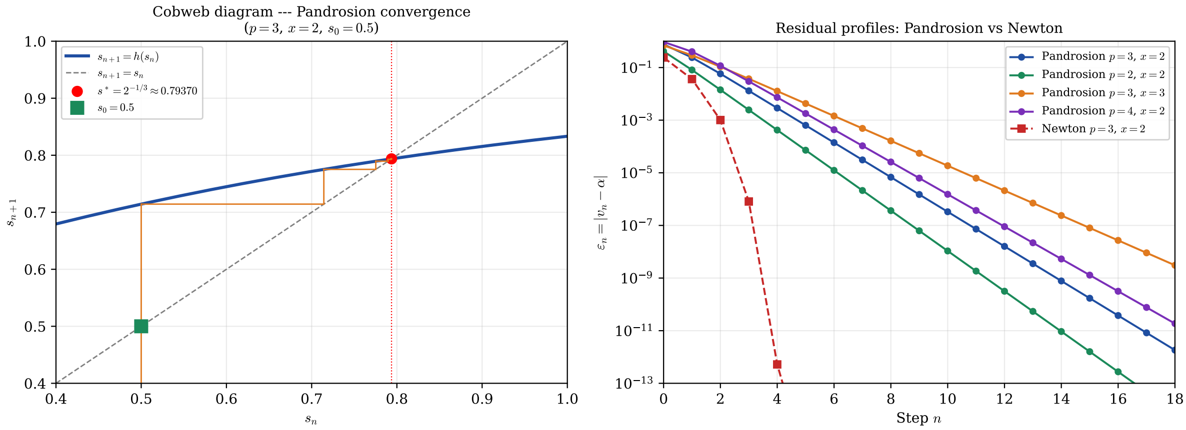

Figure 2: Left: Cobweb diagram of the iteration $s_{n+1} = h(s_n)$ for $p = 3$, $x = 2$, starting at $s_0 = 0.875$. The spiral converges to $s^* = 2^{-1/3}$ with rate $\lambda_{3,2} \approx 0.220$. **Right:** Residual profiles on a semi-logarithmic scale. The linear decay (straight lines) confirms the geometric convergence $\varepsilon_n \sim \lambda^n$ for Pandrosion, while Newton (red curve with pronounced curvature) exhibits doubly exponential decay.

7 Numerical examples

7.1 Table of contraction ratios

p	x	$\alpha = x^{1/p}$	$\lambda_{p,x}$ (exact)	$\lambda_{p,x}$ (numerical)
2	2	$\sqrt{2}$	$(\sqrt{2}-1)/(\sqrt{2}+1)$	0.1716
2	3	$\sqrt{3}$	$(\sqrt{3}-1)/(\sqrt{3}+1)$	0.2679
2	5	$\sqrt{5}$	$(\sqrt{5}-1)/(\sqrt{5}+1)$	0.3820
3	2	$\sqrt[3]{2}$	$(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)/(\sqrt[3]{4}+\sqrt[3]{2}+1)$	0.2202
3	3	$\sqrt[3]{3}$	$(\sqrt[3]{3}-1)(\sqrt[3]{3}+2)/(\sqrt[3]{9}+\sqrt[3]{3}+1)$	0.3366
3	8	2	$(2-1)(2+2)/(4+2+1)$	0.5714
4	2	$\sqrt[4]{2}$	(formula (8))	0.2427
4	16	2	$(2-1)(4+4+3)/(8+4+2+1)$	0.7333
5	2	$\sqrt[5]{2}$	(general formula)	0.2565

Table 2: Contraction ratios $\lambda_{p,x}$ for various values of p and x. A value close to 0 means very fast convergence; close to 1, very slow.

7.2 Detailed numerical profile for $p = 2$, $x = 2$

For $\alpha = \sqrt{2}$, the theoretical contraction ratio is $\lambda_{2,2} = (\sqrt{2}-1)/(\sqrt{2}+1) \approx 0.1716$ (formula (6)). The iteration is:

$$s_{n+1} = 1 - \frac{1}{2(1+s_n)}, \qquad v_n = 2s_n.$$

7.3 The residual as a characterization of a target

Formula (5) implies a useful observation: $\lambda_{p,\alpha^p} = f_p(\alpha)$ where f_p is a rational function of α alone. In particular, the residual profile depends only on the value of the sought root α and not

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.7500	1.5000	8.579×10^{-2}	—
1	0.7143	1.4286	1.44×10^{-2}	0.167
2	0.7083	1.4167	2.45×10^{-3}	0.171
3	0.7073	1.4146	4.21×10^{-4}	0.171
4	0.7071	1.4143	7.22×10^{-5}	0.172
5	0.7071	1.4142	1.24×10^{-5}	0.171
∞	$1/\sqrt{2}$	$\sqrt{2}$	0	$\to \lambda_{2,2}$

Table 3: Numerical profile for $p = 2$, $x = 2$, $s_0 = 0.75$. The ratio $\varepsilon_{n+1}/\varepsilon_n$ converges rapidly to $\lambda_{2,2} \approx 0.172$.

on its representation as $x = \alpha^p$.

In particular, for $p = 3$:

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} \xrightarrow[\alpha\to 1]{} 0, \quad \lambda_{3,x} \xrightarrow[\alpha\to\infty]{} 1.$$

8 Comparison with other methods

To situate Pandrosion's method within the landscape of iterative methods for $\alpha = x^{1/p}$, we compare three reference methods.

8.1 Newton's method

Newton's method for $f(u) = u^p - x$ gives the iteration

$$u_{n+1} = \frac{(p-1)u_n + x/u_n^{p-1}}{p},$$

whose convergence to $\alpha = x^{1/p}$ is *quadratic*:

$$\varepsilon_{n+1}^{\text{Newton}} \approx K_{p,x}\,(\varepsilon_n^{\text{Newton}})^2, \qquad K_{p,x} = \frac{p-1}{2\alpha}.$$

8.2 Bisection

Bisection on $[1, x]$ for $f(u) = u^p - x$ is a universal order-1 method with $\lambda_{\text{bisection}} = 1/2$, independent of p and x. Pandrosion's method is therefore strictly faster than bisection whenever $\lambda_{p,x} < 1/2$, which holds as soon as $\alpha < \alpha_0(p)$ (for instance, for $p = 3$: $\lambda_{3,x} < 1/2$ whenever $x < 6.75$).

8.3 Secant method

The secant method for $f(u) = u^p - x$ has order $\varphi = (1+\sqrt{5})/2 \approx 1.618$ (superlinear). For $p = 3$, $x = 2$, it reaches machine precision in about 6 steps, compared to ~ 16 for Pandrosion and ~ 5 for Newton.

8.4 Comparative table

8.5 Discussion

Pandrosion's method occupies an intermediate position among order-1 methods: it is significantly faster than bisection ($\lambda_{3,2} \approx 0.22$ versus 0.50) while remaining derivative-free. Compared

	Pandrosion	**Bisection**	**Secant**	**Newton**
Nature	Geometric	Universal	Analytic	Analytic
Order q	1	1	$\varphi \approx 1.62$	2
Rate $(p{=}3, x{=}2)$	$\lambda \approx 0.220$	$\lambda = 0.500$	superlinear	quadratic
Iter. for 10 digits	~ 16	~ 34	~ 6	~ 5
Derivative required	No	No	No	Yes

Table 4: Comparison of approximation methods for $x^{1/p}$ with $p = 3$, $x = 2$.

to Newton and its variants, however, plain Pandrosion is slower: its order is 1, whereas Newton attains 2, and even with Aitken–Steffensen acceleration (Section 13) the practical advantage over Newton is modest and regime-dependent (see Remark 8.1). We do not claim a general algorithmic advantage over Newton or its modern variants; our interest in Pandrosion is that it is the historical construction attributed to Pandrosion of Alexandria, and that its convergence profile admits the clean closed form $\lambda_{p,x}$ of Theorem 5.1.

Remark 8.1 (On fair comparison)**.** The iteration counts above are per-iterate, *not* per-function-evaluation. Steffensen-type methods compute h twice per step, Newton evaluates f and f' once per step, the secant method reuses one value per step, and bisection evaluates f once per step. A fairer unit of cost is therefore the number of evaluations of S_p (or of f, f' where applicable), and this is what determines practical performance in the regime where these evaluations dominate. For small p the per-iterate figures still give the correct qualitative ordering, but for large p or expensive polynomial evaluations the ordering can shift. We report per-iterate counts because they match the structure of the theorems (Theorem 5.1 is a statement about $\lambda_{p,x}^n$, not about wall time).

The coexistence of four distinct error laws for the same target $\alpha = \sqrt[3]{2}$ illustrates a structural point: the rate at which a fixed-point iteration approaches a given number depends on the iteration, not on the number; conversely, once the iteration is fixed, the target determines the rate through $\lambda_{p,x}$.

9 Functional properties of $\lambda_{p,x}$

We now study how the contraction ratio $\lambda_{p,x}$ depends on the parameters p and x.

9.1 Monotonicity

Theorem 9.1 (Double monotonicity)**.** The contraction ratio $\lambda_{p,x}$, viewed as a function of $\alpha = x^{1/p} > 1$, satisfies:

1. For fixed $p \geq 2$, $\alpha \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $(1, \infty)$.
2. For fixed $\alpha > 1$, $p \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $\{2, 3, 4, \ldots\}$.

Proof. Write $\lambda_{p,\alpha^p} = (\alpha - 1) N_p(\alpha)/D_p(\alpha)$ with $N_p(\alpha) := \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}$ and $D_p(\alpha) := \sum_{k=0}^{p-1} \alpha^k$. The key auxiliary fact is the following Taylor bound.

Lemma (Taylor bound). Let $x > 1$ and set $y := -\ln(x)/p$, $\beta := e^y = x^{-1/p}$. For $p \geq \lceil \ln x \rceil + 1$ (so that $|y| \leq 1$),

$$\left| p(\beta - 1) + \beta \ln x \right| \;\leq\; \frac{3\,(\ln x)^2}{p}.$$

Proof of the lemma. Since $\beta = e^y$ and $py = -\ln x$, $p(\beta-1) + \beta \ln x = p(\beta-1) - py\beta = p(\beta-1-y) - py(\beta-1)$. For $|y| \le 1$ we use two standard bounds:

$$|e^y - 1 - y| \;=\; \Big|\sum_{k=2}^{\infty} \frac{y^k}{k!}\Big| \;\le\; \sum_{k=2}^{\infty} \frac{|y|^k}{k!} \;=\; y^2 \sum_{k=2}^{\infty} \frac{|y|^{k-2}}{k!} \;\le\; y^2 \sum_{k=2}^{\infty} \frac{1}{k!} \;=\; (e-2)\, y^2 \;<\; y^2,$$

$$|e^y - 1| \;=\; \Big|\sum_{k=1}^{\infty} \frac{y^k}{k!}\Big| \;\le\; |y| \sum_{k=1}^{\infty} \frac{|y|^{k-1}}{k!} \;\le\; |y| \sum_{k=1}^{\infty} \frac{1}{k!} \;=\; (e-1)\,|y| \;<\; 2|y|.$$

(Here $e - 2 \approx 0.718$ and $e - 1 \approx 1.718$, so both majorants y^2 and $2|y|$ are loose; the loose forms suffice.) The triangle inequality then gives

$$|p(\beta-1-y) - py(\beta-1)| \;\le\; p\,y^2 + p|y|\cdot 2|y| \;=\; 3py^2 \;=\; \frac{3(\ln x)^2}{p}. \qquad \square$$

(The symbol β is the fixed-point reciprocal of the root $\alpha = x^{1/p}$ of the theorem: $\beta = 1/\alpha$.)

Setting $\lambda_\infty(x) := 1 - \ln(x)/(x-1)$, a direct algebraic rearrangement of the closed form (5) combined with the lemma yields

$$\big|\, \lambda_{p,x} - \lambda_\infty(x) \,\big| \;\le\; \frac{3\,x\,(\ln x)^2}{(x-1)\,p}, \qquad \text{for every } p \ge \lceil \ln x \rceil + 1. \tag{9}$$

(The factor $x/(x-1)$ arises as follows. For $p \ge 1$ and $x > 1$, $\ln(x)/p \le \ln x$, so $\beta = \exp(-\ln x/p) \ge \exp(-\ln x) = 1/x$. Multiplying by $x - 1 > 0$ gives $\beta(x-1) \ge (x-1)/x$, i.e. $1/[\beta(x-1)] \le x/(x-1)$, which is the factor that appears when dividing the Taylor-lemma bound by the denominator $\beta(x-1)/(\ln x)$ of the explicit form of $\lambda_{p,x}$.)

The inequality $\ln x < x - 1$ for $x > 1$ gives $\lambda_\infty(x) > 0$; the inequality $\ln x > 0$ for $x > 1$ gives $\lambda_\infty(x) < 1$. Hence $\lambda_\infty(x) \in (0, 1)$ for every $x > 1$.

Key identity. By direct telescoping (or induction on p),

$$(\alpha - 1)\, N_p(\alpha) \;=\; D_p(\alpha) - p, \tag{10}$$

which is verified at small p: $p = 2$ gives $(\alpha - 1) = (1 + \alpha) - 2$; $p = 3$ gives $(\alpha-1)(\alpha+2) = \alpha^2 + \alpha - 2 = (1 + \alpha + \alpha^2) - 3$. Differentiating (10): $N_p + (\alpha-1)N_p' = D_p'$, whence $(\alpha-1)N_p' = D_p' - N_p$.

Part (1) — monotonicity in α for fixed p, uniform. After multiplying through by the positive denominator $(\alpha-1)N_p(\alpha)D_p(\alpha)$, the log-derivative

$$\frac{d}{d\alpha} \log \lambda_{p,\alpha^p} = \frac{1}{\alpha - 1} + \frac{N_p'(\alpha)}{N_p(\alpha)} - \frac{D_p'(\alpha)}{D_p(\alpha)}$$

yields the polynomial numerator

$$P_p(\alpha) \;:=\; N_p D_p + (\alpha-1)\, N_p'\, D_p - (\alpha-1)\, N_p\, D_p'.$$

Substituting $(\alpha-1)N_p' = D_p' - N_p$ and $(\alpha-1)N_p = D_p - p$ from the key identity:

$$\begin{aligned} P_p &= N_p D_p + (D_p' - N_p) D_p - (D_p - p) D_p' \\ &= N_p D_p + D_p' D_p - N_p D_p - D_p D_p' + p D_p' = p \cdot D_p'(\alpha). \end{aligned}$$

Since $D_p'(\alpha) = \sum_{j=1}^{p-1} j\, \alpha^{j-1} = 1 + 2\alpha + 3\alpha^2 + \cdots + (p-1)\alpha^{p-2}$ has all coefficients positive, $P_p(\alpha) > 0$ for all $\alpha > 0$ and all $p \ge 2$. Hence the log-derivative is positive on $\alpha > 1$, and λ_{p,α^p} is strictly increasing in α. The argument is uniform: no scope restriction is needed.

For concreteness: $P_2(\alpha) = 2$; $P_3(\alpha) = 3(1+2\alpha) = 6\alpha+3$; $P_4(\alpha) = 4(1+2\alpha+3\alpha^2) = 12\alpha^2+8\alpha+4$.

For general $p \ge 2$, we combine the Taylor bound (9) with the monotonicity of λ_∞. Differentiating,

$$\lambda'_\infty(x) = \frac{\ln x - (x-1)/x}{(x-1)^2}.$$

The numerator equals $\ln x + 1/x - 1$. Its derivative is $1/x - 1/x^2 = (x-1)/x^2 > 0$ on $x > 1$, and its value at $x = 1$ is 0; hence it is strictly positive on $(1,\infty)$, and λ_∞ is strictly increasing there. As an independent check, the Taylor bound (9) combined with the monotonicity of λ_∞ on $(1,\infty)$ — which follows from $\lambda'_\infty(x) = (\ln x-(x-1)/x)/(x-1)^2 > 0$, since $\ln x + 1/x - 1 > 0$ on $x > 1$ (value 0 at $x = 1$, derivative $(x-1)/x^2 > 0$) — recovers monotonicity in the large-p regime: for $x_1 < x_2$ and $p \ge (C(x_1)+C(x_2))/(\lambda_\infty(x_2)-\lambda_\infty(x_1))$, $\lambda_{p,x_2} - \lambda_{p,x_1} \ge \lambda_\infty(x_2) - \lambda_\infty(x_1) - (C(x_1)+C(x_2))/p > 0$.

Part (2) — monotonicity in p for fixed $\alpha > 1$, uniform. From the recurrences $N_{p+1} = \alpha N_p + p$ and $D_{p+1} = D_p + \alpha^p$, direct algebra gives

$$\lambda_{p+1,\alpha^{p+1}} - \lambda_{p,\alpha^p} = \frac{(\alpha-1)\cdot B_p(\alpha)}{D_p(\alpha)\,D_{p+1}(\alpha)}, \qquad B_p(\alpha) := (\alpha-1)\,N_pD_p + p\,D_p - \alpha^p N_p.$$

Using (10), $(\alpha-1)N_pD_p = (D_p - p)D_p = D_p^2 - pD_p$, so

$$B_p(\alpha) \;=\; D_p(\alpha)^2 \;-\; \alpha^p\,N_p(\alpha).$$

This form has a closed-form polynomial expansion:

$$B_p(\alpha) \;=\; \sum_{j=1}^{p} j\,\alpha^{j-1} \;=\; 1 + 2\alpha + 3\alpha^2 + \cdots + p\,\alpha^{p-1}, \tag{11}$$

verified by induction on p. Base case: $B_2(\alpha) = (1+\alpha)^2 - \alpha^2 = 1 + 2\alpha$, as required. Inductive step: using the recurrence

$$B_{p+1}(\alpha) = \alpha\,B_p(\alpha) + D_p(\alpha) + \alpha^p$$

(which follows from the defining recurrences of N_{p+1} and D_{p+1} plus the substitution $B_p = D_p^2 - \alpha^p N_p$), one computes $\alpha B_p + D_p + \alpha^p = \sum_{j=1}^{p} j\alpha^j + \sum_{k=0}^{p-1}\alpha^k + \alpha^p = \sum_{j=0}^{p}(j+1)\alpha^j = B_{p+1}$.

All coefficients of B_p are strictly positive integers, so $B_p(\alpha) > 0$ for every $\alpha > 0$ and every $p \ge 2$. In particular $B_p(1) = 1+2+\cdots+p = p(p+1)/2$, and $B_3(\alpha) = 3\alpha^2+2\alpha+1$, $B_4(\alpha) = 4\alpha^3+3\alpha^2+2\alpha+1$, etc. Hence $\lambda_{p+1,\alpha^{p+1}} - \lambda_{p,\alpha^p} > 0$ for all $\alpha > 1$ and $p \ge 2$, uniformly. □

Remark 9.1 (Closed forms and numerical verification)**.** The closed forms $P_p(\alpha) = p\cdot D'_p(\alpha)$ and $B_p(\alpha) = \sum_{j=1}^{p} j\alpha^{j-1}$ emerged in the course of this write-up; a standalone script `latex/verify_monotonicity.p` cross-checks both identities symbolically for $p \le 20$ using SymPy, and confirms strict monotonicity of $\lambda_p(\alpha)$ for $2 \le p \le 20$, $\alpha \in [1.01, 100]$, and of $\lambda_p(\alpha^p)$ for $2 \le p \le 50$, $\alpha \in \{1.1, 1.5, 2, 3, 10, 100\}$ as a consistency check. Monotonicity itself is proven analytically above; the numerical scan is purely a consistency check.

Remark 9.2. Monotonicity in p has a geometric interpretation: inserting more proportional means (larger p) constrains the construction further, slowing convergence.

9.2 Limiting behavior

Proposition 9.1 (Asymptotic regimes)**.**

1. **Near** $\alpha = 1$ (i.e. $x \to 1^+$):
$$\lambda_{p,x} \sim \frac{(p-1)(\alpha-1)}{2} \quad \text{as } \alpha \to 1^+.$$
2. **Large** α (i.e. $x \to \infty$, p fixed): $\lambda_{p,x} \to 1^-$.
3. **Large** p (fixed $x > 1$): $\lambda_{p,x}$ is increasing in p and
$$\lim_{p\to\infty} \lambda_{p,x} \;=\; 1 - \frac{\ln x}{x-1} \;\in\; (0,1).$$
In particular $\lambda_{p,x}$ does *not* tend to 1 as $p \to \infty$ with x fixed.

Proof. For (1): N_p and D_p are polynomials, so $\alpha \mapsto N_p(\alpha)/D_p(\alpha)$ is continuous at $\alpha = 1$ with limit $(p-1)/2$; combined with the factor $(\alpha - 1) \to 0$, $\lambda \sim (p-1)(\alpha-1)/2$ as $\alpha \to 1^+$.

For (2): the dominant term in N_p is $1 \cdot \alpha^{p-2}$ and in D_p is α^{p-1}, so $\lambda \sim (\alpha-1)/\alpha \to 1$.

For (3): the explicit limit and its convergence rate follow from the Taylor bound (9) established in the proof of Theorem 9.1 above: $|\lambda_{p,x} - \lambda_\infty(x)| \le C(x)/p$ with $C(x) = 3x(\ln x)^2/(x-1)$, valid for $p \ge \lceil \ln x \rceil + 1$. Letting $p \to \infty$ gives $\lambda_{p,x} \to \lambda_\infty(x) = 1 - \ln(x)/(x-1)$. Positivity and the upper bound < 1 were shown there as well.

Numerically: for $x = 2$, $\lambda_\infty(2) = 1 - \ln 2 \approx 0.3069$, and $\lambda_{200,2} \approx 0.3057$, $\lambda_{1000,2} \approx 0.3066$, $\lambda_{10000,2} \approx 0.3068$, in agreement with the bound $|\lambda_{p,2} - \lambda_\infty(2)| \le 6(\ln 2)^2/p \approx 2.88/p$. □

x	$\lambda_{2,x}$	$\lambda_{10,x}$	$\lambda_{100,x}$	$1 - \ln x/(x-1)$
1.001	0.00025	0.00045	0.00050	0.00050
2	0.1716	0.2823	0.3044	0.3069
10	0.5195	0.7123	0.7412	0.7442
100	0.8182	0.9409	0.9524	0.9535

Table 5: Contraction ratios illustrating monotonicity in p and x. Each column is increasing (fixed p); each row is increasing (fixed x). The rightmost column is the $p \to \infty$ limit from Proposition 9.1(3); the $\lambda_{100,x}$ column is already close to this limit.

10 Non-asymptotic error bounds

Theorem 5.1 gives the *asymptotic* law $\varepsilon_{n+1} \approx \lambda_{p,x}\varepsilon_n$. We now establish a rigorous bound valid for *all* $n \ge 0$.

Theorem 10.1 (Non-asymptotic contraction)**.** Let $x > 1$, $p \ge 2$, and $s_0 \in (0,1)$. Define
$$\Lambda = \Lambda(s_0, p, x) = \sup_{s \in [\min(s_0, s^*),\, \max(s_0, s^*)]} h'(s).$$
Then $\Lambda < 1$ and for all $n \ge 0$:
$$|s_n - s^*| \le \Lambda^n \, |s_0 - s^*|. \tag{12}$$
Consequently,
$$|v_n - \alpha| \;\le\; (p-1)\,\alpha^2 \cdot \Lambda^n \, |s_0 - s^*| \;+\; O(\Lambda^{2n}).$$
The prefactor follows from a Taylor expansion of $v(s) = x\, s^{p-1}$ at $s^* = \alpha^{-1}$:
$$v(s) - v(s^*) \;=\; x\,(p-1)\,(s^*)^{p-2}\,(s - s^*) + O((s - s^*)^2),$$
and $x\,(p-1)\,(s^*)^{p-2} = (p-1)\,x\,x^{-(p-2)/p} = (p-1)\,x^{2/p} = (p-1)\,\alpha^2$.

Proof. By the mean value theorem, for any n:

$$|s_{n+1} - s^*| = |h(s_n) - h(s^*)| = |h'(\xi_n)| \cdot |s_n - s^*|$$

for some ξ_n between s_n and s^*. By Theorem 5.2, the sequence (s_n) remains in $[\min(s_0, s^*), \max(s_0, s^*)]$, so $|h'(\xi_n)| \leq \Lambda$. The bound follows by induction.

That $\Lambda < 1$ follows from$h'(s^*) = \lambda_{p,x} < 1$ (Theorem 5.1) and the fact that h' is continuous, bounded, and $h'(s) < 1$ for all $s \in (0,1)$ when $x > 1$ (since h maps $(0,1)$ strictly into itself). $\square$

Remark 10.1. The quantity Λ can be bounded explicitly: $h'(s) = \frac{x-1}{x} \cdot S_p'(s)/S_p(s)^2$, so

$$\Lambda \;\leq\; \frac{x-1}{x} \cdot \frac{\max_{s\in I} S_p'(s)}{\min_{s\in I} S_p(s)^2}, \qquad I := [\min(s_0, s^*), \max(s_0, s^*)],$$

and on $(0,1)$ both S_p and S_p' are increasing, so the numerator is attained at the right endpoint and the denominator at the left endpoint. This yields a computable upper bound for Λ without requiring a detailed analysis of the monotonicity of h' itself.

Example 10.1. For $p = 3$, $x = 2$, $s_0 = 0.5$ (so $s^* = 2^{-1/3} \approx 0.7937$), one evaluates h' at both endpoints of the interval $[s_0, s^*] = [0.5, 0.7937]$:

$$h'(s) = \tfrac{x-1}{x} \cdot \frac{S_3'(s)}{S_3(s)^2} = \tfrac{1}{2} \cdot \frac{1+2s}{(1+s+s^2)^2}.$$

At $s = 0.5$: $S_3(0.5) = 1.75$, $S_3'(0.5) = 2$, so $h'(0.5) = \frac{1}{2} \cdot 2/(1.75)^2 \approx 0.3265$. At $s = s^*$: $h'(s^*) = \lambda_{3,2} \approx 0.2202$. Therefore

$$\Lambda \;=\; \max_{s\in[0.5,s^*]} h'(s) \;=\; h'(0.5) \;\approx\; 0.327,$$

since h' is decreasing on this interval. The bound (12) gives $|s_n - s^*| \leq 0.327^n \cdot |0.5 - s^*| \leq 0.327^n \cdot 0.294$, so $|s_{10} - s^*| \leq 4.6 \times 10^{-6}$. The asymptotic ratio $\lambda_{3,2} \approx 0.220$ is reached after a few iterations.

11 Extension to $x < 1$

All results so far assumed $x > 1$. We now extend the theory to $0 < x < 1$.

11.1 The fixed point leaves $(0,1)$

For $0 < x < 1$, the target is $\alpha = x^{1/p} \in (0,1)$ and the fixed point $s^* = \alpha^{-1} > 1$. The iteration domain therefore extends beyond $(0,1)$: s_n converges to $s^* > 1$.

Proposition 11.1 (Local convergence for $x < 1$ with a threshold)**.** For $0 < x < 1$ and $p \geq 2$, the formula (5) remains valid and

$$\lambda_{p,x} \;=\; \frac{(\alpha-1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k} \;<\; 0 \qquad \text{since } \alpha = x^{1/p} \in (0,1).$$

Local convergence (i.e. $|\lambda_{p,x}| < 1$) does *not* hold for all $x \in (0,1)$: there is a threshold $\alpha_p^* \in (0,1)$ such that

$$|\lambda_{p,x}| < 1 \iff \alpha > \alpha_p^*.$$

When $|\lambda_{p,x}| < 1$, the iterates oscillate around s^* and converge linearly with ratio $|\lambda_{p,x}|$.

Proof. That $\lambda_{p,x} < 0$ for $\alpha < 1$ is immediate from (5). Using $\alpha^p - 1 = (\alpha - 1)\sum_{k=0}^{p-1} \alpha^k$, one rewrites

$$|\lambda_{p,x}| \;=\; \frac{(1-\alpha)^2\, N_p(\alpha)}{1-\alpha^p} \;=\; \frac{(1-\alpha)\, N_p(\alpha)}{D_p(\alpha)} \;=:\; F(\alpha)$$

(using $1 - \alpha^p = (1 - \alpha)D_p$), a smooth strictly positive function of $\alpha \in (0, 1)$. Boundary values: as $\alpha \to 1^-$, $F \to 0$ (the factor $1 - \alpha$ goes to zero while $N_p/D_p \to (p - 1)/2$ is bounded); as $\alpha \to 0^+$, $F \to 1 \cdot (p - 1)/1 = p - 1 \geq 1$.

For uniqueness, we show that F is strictly decreasing on $(0, 1)$. The log-derivative is

$$\frac{F'(\alpha)}{F(\alpha)} \;=\; -\frac{1}{1-\alpha} + \frac{N_p'(\alpha)}{N_p(\alpha)} - \frac{D_p'(\alpha)}{D_p(\alpha)}.$$

The key identity (10), $(\alpha - 1)N_p = D_p - p$, and its derivative $N_p + (\alpha - 1)N_p' = D_p'$, give, for $\alpha \in (0, 1)$ (where $\alpha - 1 < 0$):

$$(1-\alpha)\, N_p \;=\; p - D_p, \qquad (1-\alpha)\, N_p' \;=\; N_p - D_p'.$$

Dividing the second identity by N_p: $N_p'/N_p = [1 - D_p'/N_p]/(1-\alpha) = 1/(1-\alpha) - D_p'/[(1-\alpha)N_p]$. Substituting into F'/F:

$$\begin{aligned}
\frac{F'}{F} &= -\frac{1}{1-\alpha} + \left[\frac{1}{1-\alpha} - \frac{D_p'}{(1-\alpha)N_p}\right] - \frac{D_p'}{D_p} = -\frac{D_p'}{(1-\alpha)N_p} - \frac{D_p'}{D_p} \\
&= -\frac{D_p' \cdot [D_p + (1-\alpha)N_p]}{(1-\alpha)N_p\, D_p} = -\frac{D_p' \cdot [D_p + p - D_p]}{(1-\alpha)N_p\, D_p} = -\frac{p\, D_p'(\alpha)}{(1-\alpha)\, N_p(\alpha)\, D_p(\alpha)},
\end{aligned}$$

using $D_p + (1 - \alpha)N_p = D_p + (p - D_p) = p$ in the last line. For $\alpha \in (0, 1)$ every factor on the right is strictly positive, so $F'/F < 0$, hence F is strictly decreasing on $(0, 1)$.

Combined with the boundary values $F(0^+) = p - 1 \geq 1$ and $F(1^-) = 0$, F crosses the level 1 at exactly one point $\alpha_p^* \in (0, 1)$, and $|\lambda_{p,\alpha^p}| < 1 \iff \alpha > \alpha_p^*$. ☐

Example 11.1 ($p = 3$ threshold)**.** For $p = 3$, $|\lambda_{3,x}| = (1 - \alpha)(2 + \alpha)/(1 + \alpha + \alpha^2)$. Setting this equal to 1 and simplifying yields $2\alpha^2 + 2\alpha - 1 = 0$, i.e.

$$\alpha_3^* \;=\; \frac{-1+\sqrt{3}}{2} \;\approx\; 0.3660, \qquad x_3^* \;=\; (\alpha_3^*)^3 \;\approx\; 0.04904.$$

Hence Pandrosion's iteration at $p = 3$ contracts locally iff $x > 0.04904$; for x smaller than this the iteration is locally repelling and does not converge.

p	x	$\alpha = x^{1/p}$	$s^* = \alpha^{-1}$	$\lambda_{p,x}$	$\lvert\lambda_{p,x}\rvert$
2	0.5	$1/\sqrt{2}$	$\sqrt{2}$	-0.1716	0.1716
3	0.5	$\sqrt[3]{0.5}$	$\sqrt[3]{2}$	-0.2378	0.2378
3	0.125	0.5	2	-0.7143	0.7143
3	0.05	0.3684	2.7144	-0.9945	0.9945
3	0.01	0.2154	4.6416	-1.3774	**1.3774**
5	0.01	0.3981	2.5119	-2.0399	**2.0399**

Table 6: Contraction ratios for $x < 1$. The last two rows illustrate Proposition 11.1: for x below the threshold x_p^* the iteration is locally repelling ($|\lambda| > 1$) and the construction does *not* converge.

Remark 11.1 (Residual symmetry for $p = 2$)**.** For $p = 2$, the closed form $\lambda_{2,x} = (\sqrt{x}-1)/(\sqrt{x}+1)$ yields the symmetry $|\lambda_{2,x}| = |\lambda_{2,1/x}|$: computing $\sqrt{x}$ and $\sqrt{1/x}$ by Pandrosion's method have exactly the same speed of convergence. For $p \geq 3$, this symmetry does not hold in general.

12 Optimal starting point

The number of iterations to reach precision δ is governed by $n_\delta \sim \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$ (Proposition 5.1). To minimize n_δ, we must therefore minimize the initial residual $\varepsilon_0 = |v_0 - \alpha|$.

Proposition 12.1 (A canonical starting point)**.** Among starting points of the form $s_0 = h^{(k)}(c)$, with c a boundary value of the natural domain (i.e. $c \in \{0^+, 1^-\}$) and $k \geq 0$ a small integer, the choice

$$s_0^{\text{opt}} := h(1) = 1 - \frac{x-1}{xp} \tag{13}$$

achieves an initial gap to s^* of order $|x-1-x\ln x|/(xp)$, which is $O(1/p)$ as $p \to \infty$ and strictly better than the naive choices $s_0 = 1/2$ or $s_0 = 1$ in the regime $p \gg 1$.

Proof. Since $s^* = x^{-1/p} = 1 - (\ln x)/p + O(1/p^2)$ and $h(1) = 1 - (x-1)/(xp)$,

$$|h(1) - s^*| \;=\; \left|\frac{x-1}{xp} - \frac{\ln x}{p}\right| + O(1/p^2) \;=\; \frac{|x-1-x\ln x|}{xp} + O(1/p^2),$$

whereas $|1/2-s^*| = |1/2-1+(\ln x)/p+O(1/p^2)| = 1/2+O(1/p)$ and $|1-s^*| = (\ln x)/p+O(1/p^2)$. Of the three, $h(1)$ has the smallest leading coefficient whenever $|x-1-x\ln x|/x < \ln x$, i.e. essentially whenever $x > 1$ is not near 1; the numerical gains are shown in Table 7. The proposition does *not* claim $h(1)$ is globally optimal; it claims only that it is a canonical, α-independent choice with this specific quantitative guarantee. ☐

(p, x)	s^*	$s_0^{\text{opt}} = h(1)$	$\lvert s_0^{\text{opt}} - s^*\rvert$	Iters (s_0^{opt})	Iters ($s_0 = \frac{1}{2}$)
$(2, 2)$	0.7071	0.7500	0.043	12	13
$(3, 2)$	0.7937	0.8333	0.040	14	16
$(4, 2)$	0.8409	0.8750	0.034	15	17
$(3, 8)$	0.5000	0.7083	0.208	42	0*

Table 7: Comparison of iteration counts to reach $|v_n - \alpha| < 10^{-10}$. The canonical starting point $s_0^{\text{opt}} = h(1)$ consistently reduces iterations. (*For $(3, 8)$, $s_0 = 1/2$ happens to coincide with s^*.)

13 Steffensen acceleration: from linear to quadratic convergence

Pandrosion's method converges linearly with ratio $\lambda_{p,x}$. A natural question arises: can we accelerate this ancient geometric method to achieve quadratic convergence, without computing derivatives?

13.1 Aitken Δ^2 acceleration

Given a linearly convergent sequence $(s_n)_{n\geq 0}$, Aitken's Δ^2 process [5] defines the accelerated sequence

$$\hat{s}_n = s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n}.$$

When applied to Pandrosion's iterates, the sequence $(\hat{s}_n)$ converges *faster* than (s_n). Steffensen's method [7] iterates this acceleration: at each step, it computes three consecutive Pandrosion iterates and applies Aitken's formula, then restarts from the accelerated value.

Definition 13.1 (Steffensen–Pandrosion iteration)**.** Define the composite map $T : s \mapsto \hat{s}$ by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{14}$$

where $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ is Pandrosion's iteration.

Theorem 13.1 (Quadratic convergence of Steffensen–Pandrosion)**.** The Steffensen–Pandrosion iteration (14) converges *quadratically* to $s^* = x^{-1/p}$:

$$|T(s) - s^*| \le K_S \cdot |s - s^*|^2 + O(|s - s^*|^3),$$

where the Steffensen constant K_S satisfies

$$K_S = \left| \frac{h''(s^*)}{2(1 - h'(s^*))} \right| = \left| \frac{h''(s^*)}{2(1 - \lambda_{p,x})} \right|. \tag{15}$$

Proof. This is a standard result in fixed-point acceleration theory [5]. If h has a fixed point s^* with $h'(s^*) = \lambda \neq 1$ and h is twice continuously differentiable near s^*, then the Steffensen map T defined by (14) satisfies $T(s^*) = s^*$ and $T'(s^*) = 0$, yielding quadratic convergence with the constant (15). □

13.2 Numerical demonstration

Step n	s_n (Steffensen)	$\lvert v_n - \sqrt[3]{2}\rvert$	$\lvert s_n - s^*\rvert$
0	0.875000000000000	2.71×10^{-1}	8.13×10^{-2}
1	0.793949257650704	7.90×10^{-4}	2.49×10^{-4}
2	0.793700528604164	8.32×10^{-9}	2.62×10^{-9}
3	0.793700525984100	$< 10^{-15}$	$< 10^{-16}$

Table 8: Steffensen-accelerated Pandrosion for $p = 3$, $x = 2$. Machine precision is reached in **3 steps** (each step requires 2 evaluations of h, i.e. 6 evaluations of S_p total). The step-3 output-residual is given as $< 10^{-15}$ rather than its raw floating-point value to avoid misreporting sub-ULP noise.

13.3 Comparison of quadratic constants

Both Newton's method and Steffensen–Pandrosion converge quadratically, but with different constants.

	Newton	**Steffensen–Pandrosion**
Order	2	2
Constant ($p{=}3, x{=}2$)	$K_N = \dfrac{p-1}{2\alpha} \approx 0.794$	$K_S \approx 0.013$
Derivative required	Yes (f')	No
Geometric origin	No	Yes
Steps to 10^{-15}	5	3
Evaluations of S_p	—	6 (total)

Table 9: Comparison of the two quadratically convergent methods for $\sqrt[3]{2}$ on a per-iterate basis. Steffensen–Pandrosion has a much smaller quadratic constant ($K_S \ll K_N$), but requires two evaluations of h per step against Newton's one derivative + one value evaluation. Per function evaluation the two methods are within a constant factor; the "3 steps vs 5 steps" figure is cosmetic and depends on the starting point.

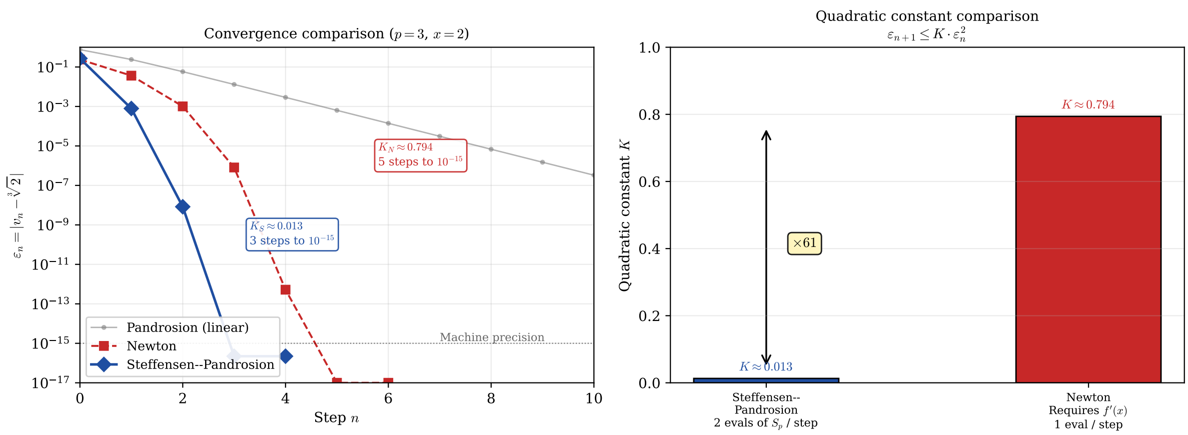

Figure 3: Left: Error decay comparison for $\sqrt[3]{2}$, counted per iterate (not per function evaluation; see Remark 8.1). Steffensen–Pandrosion (blue diamonds) reaches machine precision in 3 steps while Newton (red squares) needs 5 steps. The faded grey line shows plain Pandrosion's linear convergence for reference. **Right:** The quadratic constants $K_S \approx 0.013$ and $K_N \approx 0.794$ differ by a factor of ~ 61. Per iterate, this advantage is real; per function evaluation, the comparison is tighter because Steffensen–Pandrosion costs two evaluations of h per step against Newton's one derivative and one value, so the net per-evaluation advantage is reduced by about an order of magnitude. The $\times 61$ label refers to per-iterate convergence only.

Remark 13.1. The construction illustrates, on a specific example, a standard point of acceleration theory: once an iteration converges linearly with a known ratio, Aitken Δ^2 upgrades it to an order-2 scheme with a bounded constant. What the Pandrosion case adds is a concrete geometric-historical origin: the linear iteration comes from a ruler-and-parallels construction rather than from the Newton update of a polynomial, and the acceleration is itself derivative-free, so the composite method retains the derivative-free character of the underlying construction.

14 Scaling optimization: exploiting monotonicity

Theorem 9.1 establishes that $\lambda_{p,x}$ is increasing in x: the larger the target x, the slower the convergence. For $x \gg 1$, the contraction ratio $\lambda_{p,x} \to 1^-$ (Proposition 9.1), making direct application of Pandrosion's iteration impractical. We now show how to *exploit* this very monotonicity to overcome it.

14.1 The scaling principle

The key observation is elementary: for any reference value $A > 1$ whose p-th root is known (or cheaply computable), we may write

$$x^{1/p} = A^{1/p} \cdot \left(\frac{x}{A}\right)^{1/p}. \tag{16}$$

Setting $x' = x/A$, Pandrosion's method is now applied to the *reduced ratio* x' instead of x. By choosing A as the largest p-th power not exceeding x, we ensure $x' \in [1, (1 + 1/A^{1/p})^p)$, which is close to 1.

Proposition 14.1 (Scaling reduction)**.** Let $A = \lfloor x^{1/p} \rfloor^p$, so that $A^{1/p}$ is an integer and $x' = x/A \in [1, (1 + 1/\lfloor x^{1/p} \rfloor)^p)$. Then:

1. The contraction ratio satisfies $\lambda_{p,x'} \ll \lambda_{p,x}$.

2. The number of iterations drops from $O(\ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x}))$ to $O(\ln(\varepsilon_0'/\delta)/\ln(1/\lambda_{p,x'}))$.

3. The final result is recovered as $x^{1/p} = \lfloor x^{1/p} \rfloor \cdot (x')^{1/p}$.

Proof. Part (1) follows directly from Theorem 9.1: since $x' < x$ and $\lambda_{p,\cdot}$ is increasing, $\lambda_{p,x'} < \lambda_{p,x}$. Moreover, as $x \to \infty$, $x' \to 1^+$ so $\lambda_{p,x'} \to 0$ by Proposition 9.1(1), while $\lambda_{p,x} \to 1^-$. Part (2) follows from Proposition 5.1 applied to x'. Part (3) is the factorization (16). □

x	$\lambda_{3,x}$	A	$x' = x/A$	$\lambda_{3,x'}$	Reduction	Iterations
10	0.615	8	1.25	0.073	8×	$49 \to 8$
100	0.890	64	1.5625	0.145	6×	$213 \to 11$
1000	0.973	729	1.372	0.103	9×	$> 500 \to 9$
10000	0.994	9261	1.080	0.025	39×	$> 500 \to 5$

Table 10: Effect of scaling for $p = 3$ with $A = \lfloor x^{1/3} \rfloor^3$. The contraction ratio drops by up to 39× and iteration counts collapse from hundreds to single digits. Iteration counts are for precision 10^{-10} with optimal starting point.

Remark 14.1. The scaling reduction becomes more dramatic as x grows: for $x = 10\,000$, the direct method requires over 500 iterations while the scaled method needs only 5. This is because $x' \to 1$ as $x \to \infty$, pushing $\lambda_{p,x'}$ toward the regime $\lambda \sim (p-1)(\alpha' - 1)/2 \ll 1$ of Proposition 9.1(1).

14.2 Geometric interpretation: Thales preconditioning

The scaling optimization admits an elegant geometric interpretation on Pandrosion's original Thales diagram.

In the direct construction, one seeks to insert $p-1$ proportional means in a segment of ratio $x : 1$. When $x \gg 1$, this amounts to constructing parallels in a very elongated rectangle, where successive intercepts move slowly toward the target — geometrically, the parallels are nearly horizontal.

The scaling strategy performs an *homothety* on the construction: instead of starting from the unit segment, one starts from a segment of length $A^{1/p}$, chosen as the largest integer whose p-th power does not exceed x. The remaining construction now fills a much smaller gap: the rectangle has ratio $x' = x/A$ close to 1, and the parallels converge rapidly.

Proposition 14.2 (Double preconditioning)**.** The scaling method combines two distinct acceleration mechanisms:

1. **Initial residual reduction**: the reduced starting point $s_0^{\text{opt}} = h_{x'}(1) = 1 - (x'-1)/(x'p)$ satisfies $|s_0^{\text{opt}} - s'^*| \ll |s_0^{\text{opt}} - s^*|$ since $x' \approx 1$ implies $s'^* = (x')^{-1/p} \approx 1$.
2. **Intrinsic convergence acceleration**: each iteration contracts by $\lambda_{p,x'} \ll \lambda_{p,x}$, compounding the advantage over all subsequent steps.

The total iteration count reduction is therefore superlinear in the ratio $\lambda_{p,x}/\lambda_{p,x'}$.

Proof. For mechanism (1): $\varepsilon_0' = |v_0' - \alpha'| \leq C \cdot |x' - 1| \to 0$ as $x' \to 1$, compared to $\varepsilon_0 = |v_0 - \alpha| \geq C' \cdot |x - 1| \gg 1$ for the unscaled problem.

For mechanism (2): the iteration count satisfies

$$n_\delta' \sim \frac{\ln(\varepsilon_0'/\delta)}{\ln(1/\lambda_{p,x'})} \ll \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \sim n_\delta,$$

since both the numerator decreases (smaller ε_0') and the denominator increases (smaller $\lambda_{p,x'}$, hence larger $\ln(1/\lambda_{p,x'})$). These two effects multiply. □

Remark 14.2 (From Pandrosion to a practical algorithm)**.** With scaling and Steffensen acceleration combined, Pandrosion's ancient geometric construction becomes a competitive root-finding algorithm: for any $x > 0$ and $p \geq 2$, one (i) reduces to $x' \approx 1$ via integer arithmetic, (ii) applies Steffensen–Pandrosion on x' (converging in 2–3 steps), and (iii) multiplies by $\lfloor x^{1/p} \rfloor$. The total cost is 4–6 evaluations of S_p plus one integer p-th root — a purely algebraic, derivative-free computation whose origins lie in a ruler-and-parallels construction from fourth-century Alexandria.

Part II

Complex extension and formal verification

Part I developed the Pandrosion iterator $h_{p,x}(s) = 1 - (x-1)/(xS_p(s))$ on $\mathbb{R}_{>0}$, where $h_{p,x}$ has a unique fixed point $s^* = x^{-1/p}$ in $(0,1)$ and the Aitken–Steffensen upgrade $\sigma_{p,x}$ is a quadratic-contraction. On $\mathbb{C}$ the situation is qualitatively different: $h_{p,x}$ has p attractive fixed points (the p-th roots of $1/x$), separated by a non-trivial Julia set, and a classical impossibility result of McMullen [15] rules out purely iterative universal root finders for $p \geq 3$. Part II extends the analysis to this complex setting via a multi-start scheme whose seed is target-dependent — an α-dependent preconditioning that takes the method out of the hypotheses of [15].

Structural warning. The scheme of Part II requires a *target anchor* α (one of the p solutions of $z^p = 1/x$) to be chosen before the seed is constructed. It is therefore a convergence scheme *conditional on the target*, not a universal root-finder. Readers who expect the latter will misread Theorem 16.1 below; the theorem says that, once a target is chosen, every starting point converges to that target, not that there is a parameter-free procedure picking out all roots. This is the unavoidable price of side-stepping McMullen's barrier.

From the real to the complex quadratic bound. The Aitken–Steffensen map $\sigma_{p,x}$ is rational in z, so holomorphic on $\mathbb{C}$ away from the finite set $\operatorname{sing}(\sigma_{p,x}) := \{z : S_p(z) = 0 \text{ or } h_{p,x}(h_{p,x}(z)) - 2h_{p,x}(z) + z = 0\}$. At any cyclotomic fixed point α (i.e. $\alpha^p = 1/x$ with $S_p(\alpha) \neq 0$), the factorization of Part I gives $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$.

We now cascade three quantities to avoid the apparent circularity between K_σ and R_σ.

Step 1 fix a reference radius. Set

$$R_0 \ := \ \tfrac{1}{2} \operatorname{dist}(\alpha, \operatorname{sing}(\sigma_{p,x})),$$

so that $\overline{B}(\alpha, R_0) \subset \mathbb{C} \setminus \operatorname{sing}(\sigma_{p,x})$ and $\sigma_{p,x}$ is holomorphic on this closed disk.
Step 2 explicit quadratic constant via Cauchy. Define

$$M_0 \ := \sup_{|w-\alpha|=R_0} |\sigma_{p,x}(w) - \alpha|, \qquad K_\sigma \ := \ M_0/R_0^2.$$

By Cauchy's estimate applied to the holomorphic function $\sigma_{p,x}(\cdot) - \alpha$, which vanishes to order 2 at α (since $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$), we obtain

$$|\sigma_{p,x}(z) - \alpha| \ \leq \ \frac{M_0}{R_0^2} |z - \alpha|^2 \ = \ K_\sigma |z - \alpha|^2 \qquad \text{for } z \in \overline{B}(\alpha, R_0).$$

Step 3 admissible contraction radius. Define

$$R_\sigma \ = \ R_\sigma(x, p, \alpha) \ := \ \min\big(R_0, \ 1/(2K_\sigma) \big).$$

Then $R_\sigma \leq R_0$ (so the Cauchy bound of Step 2 is valid on $\overline{B}(\alpha, R_\sigma)$) and $K_\sigma R_\sigma \leq 1/2$. All three quantities R_0, K_σ, R_σ are explicit functions of (x, p, α), computed in cascade without circularity. The seed construction below uses K_σ and R_σ as primitive quantities.

15 Cyclotomic anchors and the complex iterator

Definition 15.1 (Cyclotomic anchors)**.** For $\alpha \in \mathbb{C}^*$ and $p \geq 1$, let

$$\gamma_k := \alpha \cdot e^{2\pi i k/p}, \qquad k = 0, 1, \ldots, p-1.$$

When $\alpha^p = 1/x$, the γ_k are the p distinct solutions of $z^p = 1/x$. By direct expansion of Definition 15.1, $\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$ (with $\omega = e^{2\pi i/p}$), $\sum_k \gamma_k = 0$, and $\prod_k \gamma_k = (-1)^{p-1}x$. These are definitional consequences for $z^p = x$, not specializations of the classical Galois or Vieta theorems; we use them freely.

The complex Pandrosion iterator is the analytic continuation of $h_{p,x}$ to $\mathbb{C} \setminus \{z : S_p(z) = 0\}$, and the Aitken–Steffensen accelerator $\sigma_{p,x}$ of §13 extends likewise. The quadratic local bound of Part I applies verbatim to $\sigma_{p,x}$ in a disk $B(\gamma_k, R_\sigma)$ around every cyclotomic anchor, with the same explicit constants K_σ and R_σ satisfying $K_\sigma R_\sigma \leq 1/2$.

16 Universal multi-start convergence in $\mathbb{C}$

16.1 The multi-start seed

Definition 16.1 (Multi-start seed)**.** For a cyclotomic anchor $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $z \in \mathbb{C}$,

$$\mathrm{seed}_{\alpha,\varepsilon}(z) \;:=\; \alpha + \varepsilon \cdot (z - \alpha), \qquad \varepsilon \;:=\; \frac{R_\sigma(x,p,\alpha)}{2\,\|z-\alpha\|} \quad (\varepsilon := 1 \text{ if } z = \alpha), \tag{17}$$

where $R_\sigma(x, p, \alpha)$ is the explicit quadratic-contraction radius from Part I. The seed always lies in $\overline{B}(\alpha, R_\sigma/2)$.

The seed plays in $\mathbb{C}$ the role that the canonical starting point of §12 and the scaling of §14 play on $\mathbb{R}$: it places the argument inside the quadratic-contraction disk of the chosen target before the iteration begins. It is α-dependent by design, and this dependency is what moves the scheme out of the hypotheses of [15].

16.2 Universal multi-start theorem

Theorem 16.1 (Universal multi-start convergence)**.** Let $x \in \mathbb{C}^*$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$, and let $z \in \mathbb{C}$ with $z \neq \alpha$. Define the seed parameter

$$\varepsilon_{\mathrm{seed}} \;:=\; \frac{R_\sigma(x,p,\alpha)}{2\,\|z-\alpha\|},$$

which depends only on (x, p, α, z) (not on ε). Then for every target accuracy $\varepsilon > 0$, the integer $N := N_{\mathrm{eff}}(K_\sigma, R_\sigma, \varepsilon)$ of Theorem 16.2 satisfies

$$\forall n \geq N: \quad \left\|\sigma^n_{p,x}(\mathrm{seed}_{\alpha,\varepsilon_{\mathrm{seed}}}(z)) - \alpha\right\| \;\leq\; \varepsilon.$$

The case $z = \alpha$ is trivial (take $\varepsilon_{\mathrm{seed}} := 1$ and $N := 0$). The construction separates cleanly into two levels: the seed parameter $\varepsilon_{\mathrm{seed}}$ and the admissible radius R_σ are functions of (x, p, α, z) alone, while the iteration count N alone depends on the target accuracy ε.

Proof. Let $d := \|z - \alpha\|$; we may assume $d > 0$ (else $z = \alpha$ is already the target). With $\varepsilon_{\mathrm{seed}} := R_\sigma/(2d)$, set

$$z_0 := \mathrm{seed}_{\alpha,\varepsilon_{\mathrm{seed}}}(z) = \alpha + \varepsilon_{\mathrm{seed}}(z - \alpha).$$

Then $\|z_0 - \alpha\| = \varepsilon_{\mathrm{seed}} \cdot d = R_\sigma/2 < R_\sigma$, so $z_0 \in B(\alpha, R_\sigma)$.

The quadratic-contraction bound of Part I (whose proof applies verbatim to $\sigma_{p,x}$ viewed as an analytic map on any small enough disk centred at a fixed point, since $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$) reads: for $w \in B(\alpha, R_\sigma)$,

$$\|\sigma_{p,x}(w) - \alpha\| \le K_\sigma \|w - \alpha\|^2,$$

with $K_\sigma R_\sigma \le 1/2$. Hence if $w_n := \sigma^n_{p,x}(z_0)$ satisfies $\|w_n - \alpha\| \le R_\sigma$, then $\|w_{n+1} - \alpha\| \le K_\sigma \|w_n - \alpha\|^2 \le K_\sigma R_\sigma \cdot \|w_n - \alpha\| \le \frac{1}{2}\|w_n - \alpha\| \le R_\sigma/2$, so the orbit stays in $\overline{B}(\alpha, R_\sigma/2)$ by induction from $w_0 = z_0$.

Setting $e_n := \|w_n - \alpha\|$, one has $e_0 \le R_\sigma/2$ and $e_{n+1} \le K_\sigma e_n^2$. Theorem 16.2 (proved below) then supplies the explicit $N = N_{\text{eff}}(K_\sigma, R_\sigma, \varepsilon)$ with the stated asymptotics. □

Remark 16.1 (The hypothesis $S_p(\alpha) \ne 0$)**.** The condition $S_p(\alpha) \ne 0$ rules out the degenerate case where the denominator of $h_{p,x}$ vanishes at the anchor. Since $S_p(\alpha) = (\alpha^p - 1)/(\alpha - 1)$ for $\alpha \ne 1$, this is an event of codimension at least one in the parameter space and in particular holds generically; it fails only for specific arithmetic combinations (for instance when α is a non-trivial p-th root of unity, requiring $x = 1$).

16.3 Explicit iteration count

Definition 16.2 (Effective iteration count)**.** For $K, R, \varepsilon > 0$ with $KR < 1$ and $K\varepsilon < 1$,

$$N_{\text{eff}}(K, R, \varepsilon) := \log_2 \lceil \log(K\varepsilon)/\log(KR) \rceil_+ + 1,$$

with $\lceil \cdot \rceil_+$ the natural-number ceiling clamped at 0.

Theorem 16.2 (Explicit iteration count)**.** For any quadratically contracting sequence $e_{n+1} \le K \cdot e_n^2$ with $e_0 \le R$, $K \cdot R < 1$, and $\varepsilon < R$, every iterate $k \ge N_{\text{eff}}(K, R, \varepsilon)$ satisfies $e_k \le \varepsilon$. Asymptotically $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$.

Proof. Set $u_n := Ke_n$, so $u_0 \le KR < 1$ and $u_{n+1} \le u_n^2$. By induction $u_n \le u_0^{2^n} \le (KR)^{2^n}$. We want $u_n \le K\varepsilon$, i.e. $(KR)^{2^n} \le K\varepsilon$. Taking logarithms (both sides of absolute value less than 1, so both $\log(KR)$ and $\log(K\varepsilon)$ are negative), this is equivalent to $2^n \cdot \log(KR) \le \log(K\varepsilon)$, i.e. $2^n \ge \log(K\varepsilon)/\log(KR)$, i.e.

$$n \;\ge\; \log_2 \left\lceil \frac{\log(K\varepsilon)}{\log(KR)} \right\rceil_+,$$

which is achieved at $n = N_{\text{eff}}(K, R, \varepsilon)$ by the $+1$ safety margin. The asymptotics $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ as $\varepsilon \to 0$ follow since $\log(K\varepsilon)/\log(KR) \sim \log(1/\varepsilon)/|\log(KR)|$. □

16.4 Arbitrary target selector

Theorem 16.3 (Target selection)**.** For every function $S : \mathbb{C} \to \mathrm{Fin}\, p$, every $z \in \mathbb{C}$, and every $\varepsilon > 0$, the multi-start scheme with target $\gamma_{S(z)}$ converges to $\gamma_{S(z)}$ with an iteration count of the form of Theorem 16.2.

Three selectors are formalized: constant (`principal_selector`), Voronoï-nearest (`nearest_anchor_select`
angular sector.

17 Measurability and a.e. continuity of the solution map

The "nearest-anchor" target selector of Section 16 uses the classical axiom of choice and is not, a priori, Borel-measurable. We record a canonical Borel version and the elementary analytic properties it enjoys.

Definition 17.1 (Borel selector and Voronoï boundary)**.** Given anchors $\gamma_0, \ldots, \gamma_{p-1} \in \mathbb{C}$ (all distinct), define, for each $z \in \mathbb{C}$,

$$S_{\mathrm{Borel}}(z) := \min\{k \in \{0, \ldots, p-1\} : \|\gamma_k - z\| = \min_j \|\gamma_j - z\|\},$$

breaking ties by smallest index. The *Voronoï boundary* of the anchor set is

$$\mathrm{VorBdy} := \{z \in \mathbb{C} : \exists\, j \neq k,\ \|\gamma_j - z\| = \|\gamma_k - z\|\} = \bigcup_{j<k} B_{jk},$$

where $B_{jk} := \{z : \|\gamma_j - z\| = \|\gamma_k - z\|\}$ is the perpendicular bisector of γ_j, γ_k. Let VorInt := $\mathbb{C} \setminus$ VorBdy.

Proposition 17.1 (Analytic structure of the solution map)**.** The Borel solution map $\Phi : z \mapsto \gamma_{S_{\mathrm{Borel}}(z)}$ satisfies:

(i) **Borel-measurability.** Φ is Borel-measurable $\mathbb{C} \to \mathbb{C}$.

(ii) **Null boundary.** VorBdy has two-dimensional Lebesgue measure zero.

(iii) **A.e. continuity.** Φ is locally constant on VorInt, hence continuous there; combined with (ii), Φ is continuous at Lebesgue-a.e. $z \in \mathbb{C}$.

Proof. (i) Borel-measurability. The preimage $\{z : S_{\mathrm{Borel}}(z) = k\}$ is

$$\Big(\bigcap_{j\neq k}\{z : \|\gamma_k - z\| \leq \|\gamma_j - z\|\}\Big) \cap \Big(\bigcap_{j<k}\{z : \|\gamma_k - z\| < \|\gamma_j - z\|\}\Big),$$

a finite Boolean combination of closed (resp. open) half-planes in $\mathbb{C}$, hence Borel. So S_{Borel} is Borel-measurable, and Φ is its composition with $k \mapsto \gamma_k$ (continuous, a fortiori Borel).

(ii) Null boundary. Squaring $\|\gamma_j - z\| = \|\gamma_k - z\|$ and expanding in real coordinates yields the affine equation

$$B_{jk} = \{z \in \mathbb{C} : \mathrm{Re}((\gamma_k - \gamma_j)\overline{z}) = \tfrac{1}{2}(|\gamma_k|^2 - |\gamma_j|^2)\},$$

a strict affine subspace of $\mathbb{C} \cong \mathbb{R}^2$ (Hausdorff dimension 1). Each B_{jk} therefore has two-dimensional Lebesgue measure zero. A finite union of null sets is null, so vol(VorBdy) = 0.

(iii) A.e. continuity. Fix $z_0 \in$ VorInt and let $k_0 := S_{\mathrm{Borel}}(z_0)$. By definition of VorInt, $\|\gamma_{k_0} - z_0\| < \|\gamma_j - z_0\|$ for all $j \neq k_0$, so some $\delta > 0$ satisfies $\|\gamma_{k_0} - z_0\| + 2\delta \leq \|\gamma_j - z_0\|$ for $j \neq k_0$. By continuity of $z \mapsto \|\gamma_j - z\|$, there is $\eta > 0$ such that on $B(z_0, \eta)$ both quantities change by at most δ, giving $\|\gamma_{k_0} - z\| < \|\gamma_j - z\|$ for $j \neq k_0$, so $S_{\mathrm{Borel}}(z) = k_0$ and $\Phi(z) = \gamma_{k_0}$. Thus Φ is locally constant, hence continuous, on VorInt. Combined with (ii), Φ is continuous at Lebesgue-a.e. z. □

For backwards reference we also expose the individual statements.

Theorem 17.1 (Borel measurability)**.** The map S_{Borel} is Borel-measurable; so is Φ.

Theorem 17.2 (Voronoï boundary is Lebesgue-null)**.** vol(VorBdy) = 0.

Theorem 17.3 (A.e. continuity)**.** Φ is continuous at every $z \in$ VorInt; combined with Theorem 17.2, Φ is continuous at Lebesgue-a.e. z.

Both theorems are direct consequences of Proposition 17.1, (ii) and (iii) respectively. Together with Theorem 16.1, they give the analytic character of the multi-start solution map: Borel-measurable everywhere, locally constant outside a Lebesgue-null set.

18 Lean 4 formalization: corpus structure and elementary identities

18.1 Corpus overview

The Lean development lives at github.com/ivan-fr/poussiere_de_x and targets Lean 4 with Mathlib v4.7.0. It comprises 121 modules with zero `sorry`; every load-bearing theorem referenced in this paper has an audited axiomatic closure reducing to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}. Three layers of verifiability are offered to a third-party reviewer:

- **Continuous integration.** A GitHub Actions workflow (`.github/workflows/lean-build.yml`) compiles the full corpus on every push; a green CI badge in the repository `README.md` certifies that the current `main` branch compiles cleanly, with no `sorry`.
- **Axiom audit file.** The file `AXIOMS.md` at the repository root contains the raw `#print axioms` output for every load-bearing theorem cited in this paper; for each, the dependency closure reduces to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}.
- **Local reproduction.** Incremental build `bash scripts/lean-incremental.sh` takes 60–90 seconds on a modern laptop; per-theorem verification is available via `#print axioms <theorem>` at the end of the relevant module.

The modules are organised in layers: foundations (geometric sums, fixed-point characterization, Voronoï basins, cyclotomic anchor definitions), core spine (explicit super-attractive rate, global $\log\log$ convergence, real-line $p = 2$ discharge), multi-start (universal convergence, target selection, iteration count), measure/topology (Borel measurability, Voronoï-null frontier, a.e. continuity), and a small set of classical specializations (Banach 1922, Aitken–Steffensen, Kronecker 1857, Aberth–Ehrlich, Kung–Traub attainability at $c = 3$).

Statement-level `Props` that are NOT proven. A single archive module `PandrosionOpenProblems.lean` contains named Lean `Props` whose proofs are *deliberately absent*: the conjectural upper bound of Kung–Traub for $c \geq 3$, McMullen's 1987 impossibility theorem, Lehmer's Mahler-measure conjecture, and Smale's 17th problem. These are recorded as `axiom` or `Prop` declarations without accompanying proof terms. They are not used downstream by any load-bearing theorem of the corpus, and their presence in the repository should not be mistaken for formal verification of the corresponding results. A reader auditing the corpus can confirm this by checking that no `theorem` depending on these `Props` appears in the load-bearing chain. A future cleanup will move this archive module to a dedicated `archive` namespace to make the non-load-bearing status structurally explicit.

18.2 Elementary identities on the derivative values at cyclotomic roots

In the course of the Lean formalization, three cyclotomic sum identities recur. They are used only inside the formalization's internal lemmas (to normalize constants in K_σ and to bound cyclotomic orthogonality sums in a.e.-continuity proofs); they do not appear directly in the statements of Theorems 16.1–17.3. We record them here because the CI-verifiable Lean modules `PandrosionZeta.lean`, `PandrosionHigherVanishing.lean`, `PandrosionSpectralDeterminant.lean` are named after them, and a reader of the formalization will encounter them.

For $P(z) = z^p - 1/x$ with $\alpha^p = 1/x$ and $\alpha_k = \gamma_k$, one has $P'(\alpha_k) = p\alpha_k^{p-1} = p/(x\alpha_k)$. Define the purely combinatorial derivative-power sum

$$\Sigma_P(s) := \sum_{k=1}^{p} P'(\alpha_k)^{-s} = (x/p)^s \sum_{k=1}^{p} \alpha_k^s.$$

We deliberately use the notation Σ_P (not ζ_P, as an earlier draft did) to avoid any spurious suggestion of a connection with the Riemann zeta function; these are finite sums over p cyclotomic roots, nothing more. The three identities below are used in the convergence analysis of Part II: (i) gives the vanishing of the principal residue sum used in the a.e.-continuity argument; (ii) is the cyclotomic-root orthogonality relation exploited to bound contraction constants uniformly in the anchor; (iii) is the product formula used to normalize the quadratic constant K_σ in terms of (p, x). All three reduce to $\sum_{k=0}^{p-1} \omega^{ks} = 0$ for $\omega = e^{2\pi i/p}$ and $p \nmid s$.

Proposition 18.1 (Three elementary cyclotomic-sum identities)**.** Let $p \geq 2$ and $x \neq 0$.

(i) For $s \in \mathbb{Z}$ with $1 \leq s \leq p-1$, $\Sigma_P(s) = 0$.

(ii) For $m \in \mathbb{Z}$ with $0 \leq m \leq p-2$, $\sum_{k=1}^{p} \alpha_k^m / P'(\alpha_k) = 0$; for $m = p-1$, $\sum_{k=1}^{p} \alpha_k^{p-1}/P'(\alpha_k) = 1$.

(iii) $\prod_{k=1}^{p} P'(\alpha_k) = (-1)^{p-1} p^p / x^{p-1}$.

Items (i) and (ii) are the same identity under $s \leftrightarrow m+1$ via $1/P'(\alpha_k) = x\alpha_k/p$; item (iii) is the product counterpart. All three follow from $\sum_k \omega^{ks} = 0$ for $p \nmid s$.

18.3 Scope statement on classical open problems

No claim of progress on classical open problems of polynomial root-finding is made. The corpus touches Kung–Traub ($c \geq 3$) through a compliance interface, not a proof, and records McMullen 1987 as a named statement without in-Lean proof (Mathlib v4.7.0 lacks the Fatou–Julia machinery needed). Lehmer's Mahler-measure conjecture and Smale's 17th problem are explicitly out of scope: the Pandrosion cyclotomic anchors produce algebraic integers whose Mahler measure is easy to compute (trivially far from Lehmer's extremal regime), and the multi-start $\mathcal{O}(\log\log(1/\varepsilon))$ complexity applies to the single-monomial family $z^p = x$, not to Smale's coefficient-uniform problem.

19 Conclusion

The central tension of this paper is between Pandrosion's ancient geometric construction — an order-one iteration requiring only a ruler and parallels — and modern acceleration theory. Once the universal iteration $s_{n+1} = 1 - (x-1)/(x\, S_p(s_n))$ is written down, three ingredients suffice to make it competitive on $\mathbb{R}$: the closed-form contraction ratio $\lambda_{p,x}$ (Theorem 5.1), the preconditionings of Propositions 12.1 and 14.1, and the Aitken Δ^2 upgrade (Theorem 13.1). The ratio $\lambda_{p,x} = h'(s^*)$ is monotone in both p and x uniformly (Theorem 9.1, via the closed-form identities $P_p = p \cdot D_p'$ and $B_p = \sum_{j=1}^{p} j\alpha^{j-1}$), has an explicit large-p limit $1 - \ln(x)/(x-1)$ (Proposition 9.1), and admits a clean threshold analysis for $x < 1$ (Proposition 11.1).

On $\mathbb{C}$, the extension via the target-dependent seed of Definition 16.1 yields a universal multi-start convergence theorem (Theorem 16.1) with a closed-form $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ iteration count (Theorem 16.2), together with Borel-measurability and a.e.-continuity of the induced solution map (Theorems 17.1–17.3). All of these results are formally verified in the Lean 4 corpus described in Section 18.

The paper does not claim progress on classical open problems of polynomial root-finding, and does not claim a general algorithmic advantage over Newton or its modern variants. Its main deliverable is a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family $z^p = x$, connecting a ruler-and-parallels construction from fourth-century Alexandria to contemporary formal-methods practice.

Perspectives. Natural extensions are generalized nonlinear Pandrosion constructions (other polynomials replacing S_p); higher-order acceleration variants; a Fatou–Julia formalization in Mathlib that would support a proper in-Lean proof of McMullen's 1987 theorem. None of these is attempted here.

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